EXP NO:15 EXPERIMENT TO UNDERSTAND THE DATA PREPROCESSING IN DATA SCIENCE

CODE :

```
import numpy as np

import pandas as pd

df = pd.read_csv("pre-process_datasample.csv")

df["Country"] = df["Country"].fillna(df["Country"].mode()[0])

df["Age"] = df["Age"].fillna(df["Age"].median())

df["Salary"] = df["Salary"].fillna(round(df["Salary"].mean()))

dummies = pd.get_dummies(df["Country"])

updated_dataset = pd.concat([dummies, df.iloc[:, [1, 2, 3]]], axis=1)

updated_dataset["Purchased"] =
updated_dataset["Purchased"].replace(["No", "Yes"], [0, 1])

print(updated_dataset)
```

OUTPUT :

Country Age Salary Purchased 0 France 44.0 72000.0 No 1 Spain 27.0 48000.0 2 Germany 30.0 54000.0 3 Spain 38.0 61000.0 Yes No No 4 Germany 40.0 NaN Yes 5 France 35.0 58000.0 Yes 6 Spain NaN 52000.0 7 France 48.0 79000.0 8 Germany 50.0 83000.0 9 France 37.0 67000.0 No Yes No Yes <class 'pandas.core.frame.DataFrame'> RangeIndex: 10 entries, 0 to 9 Data columns (total 4 columns): # Column Non-Null Count Dtype --- ------ -------------- ----- 0 Country 10 non-null object 1 Age 9 non-null float64 2 Salary 9 non-null float64 3 Purchased 10 non-null object dtypes: float64(2),

object(2) memory usage: 452.0+ bytes None 0 France Name: Country, dtype: object France <class 'pandas.core.series.Series'> Country Age Salary Purchased 0 France 44.0 72000.0 1 Spain 27.0 48000.0 No Yes 2 Germany 30.0 54000.0 3 Spain 38.0 61000.0 4 Germany 40.0 63778.0 No No Yes 5 France 35.0 58000.0 6 Spain 38.0 52000.0 7 France 48.0 79000.0 8 Germany 50.0 83000.0 9 France 37.0 67000.0 France Germany Spain 0 True False False 1 False False True 2 False True False 3 False False True 4 False True False 5 True False False 6 False False True 7 True False False 8 False True False 9 True False False Yes No Yes No Yes France Germany Spain Age Salary Purchased 0 True False False 44.0 72000.0 No 1 False False True 27.0 48000.0 Yes 2 False True False 30.0 54000.0 No 3 False False True 38.0 61000.0 No 4 False True False 40.0 63778.0 Yes 5 True False False 35.0 58000.0 Yes 6 False False True 38.0 52000.0 No 7 True False False 48.0 79000.0 Yes 8 False True False 50.0 83000.0 No 9 True False False 37.0 67000.0 Yes <class 'pandas.core.frame.DataFrame'> RangeIndex: 10 entries, 0 to 9 Data columns (total 4 columns): # Column Non-Null Count Dtype --- ------ -------------- ----- 0 Country 10 non-null object 1 Age 10 non-null float64 2 Salary 10 non-null float64 3 Purchased 10 non-null object dtypes: float64(2), object(2) memory usage: 452.0+ bytes None France Germany Spain Age Salary Purchased 0 True False False 44.0 72000.0 1 False False True 27.0 48000.0 2 False True False 30.0 54000.0 3 False False True 38.0 61000.0 0 1 0 0 4 False True False 40.0 63778.0 1 5 True False False 35.0 58000.0 1 6 False False True 38.0 52000.0 0 7 True False False 48.0 79000.0

1 8 False True False 50.0 83000.0 0 9 True False False 37.0 67000.0 1